

Technical Skills

Languages	C#, Kotlin, Java, TypeScript, JavaScript
Libraries / Frameworks	.NET Core, Entity Framework, FastEndpoints, Angular, React, Node, Web API, FluentValidation, xUnit, Testcontainers, Dapper
Databases	PostgreSQL, MongoDB, SQL Server, DynamoDB, CosmosDB
Systems / Platforms	GCP, AWS, Azure, Kubernetes, Docker, Pulumi, Terraform

Work Experience

Feb '26 - Present	Senior Software Engineer - Duffel Travel Fintech Platform - Designed and built Duffel's profit-share system end-to-end in Elixir: a tiered pricing calculation engine with FX normalisation across multiple currencies, Ecto schemas with optimistic locking, 10+ Phoenix LiveView admin pages, CLI operational tooling, and 5 production database migrations - Built the supplier commission estimates pipeline from scratch using a cross-vertical contract pattern with Elixir behaviours, Oban background job ingestion with fan-out processing, and conflict-aware upsert logic for idempotent data ingestion - Built a virtual card reconciliation platform in Phoenix LiveView for Conferma supplier cards spanning 5 umbrella apps: case management with comments, manual reconciliation workflows, automatic card ID resolution for refunds, and revenue discrepancy creation - Engineered CSV commission ingestion pipelines for three hotel suppliers (Booking.com, Onyx, Priceline), handling real-world data quality issues including UTF-8 BOM encoding, inconsistent field casing, truncated timestamps, and nullable fields - Managed zero-downtime schema migrations in a high-availability deployment, executing a full add-column, backfill, drop-legacy lifecycle for the tokenised cards feature across 9 umbrella apps and 21 files - Fixed a production timeout in draft payout creation for high-volume organisations by optimising the underlying Ecto query, and built 5 CLI tasks enabling the payments team to self-serve profit-share runs, payout management, and commission data corrections
Nov '25 - Present	Founder / CTO - Wavara Cycle-Aware Weight Tracking App - Designed and built a full-stack polyglot monorepo from scratch — a React Native mobile app (Expo SDK 54, TypeScript, React 19) paired with a C# .NET 10 Web API, shipping to both iOS and Android - Implemented a local-first architecture using expo-sqlite for offline weight and cycle tracking, with a migration-based schema and repository pattern, designed for future backend sync - Built a vertical-slice API using FastEndpoints (REPR pattern) with PostgreSQL 17, Entity Framework Core 10, JWT bearer authentication, email verification flows, and rate-limited anonymous endpoints - Engineered a fully bespoke charting library from scratch using D3 and React Native SVG, with cycle-phase-coloured backgrounds, selectable date ranges, and tap-to-inspect tooltips - Developed a bespoke design system with Reanimated 4 spring and timing animations, @shopify/react-native-skia particle effects, and a warm-dark colour palette with phase-based visual hierarchy - Established a type-safe API contract workflow — OpenAPI spec generated from the .NET API, consumed via openapi-typescript codegen and openapi-fetch in the mobile app - Deployed the API to Hetzner Cloud (ARM64) with Docker Compose, Caddy for automatic TLS, and Resend for transactional email — production and dev environments on a single server - Configured EAS Build pipelines for development, preview, and production profiles with auto-versioning, Sentry error tracking, and consent-gated PostHog product analytics - Wrote tests across both stacks: Jest + React Native Testing Library for the mobile app; xUnit + Testcontainers (throwaway PostgreSQL) for API integration tests - Built the cycle-phase calculation engine that determines the user's current menstrual phase from period start dates and configurable cycle length, driving phase-coloured charts, personalised daily guidance, and the app's visual hierarchy
Jun '24 -	Software Engineer - Propelle Ltd. Wealth Management Platform

- Architected and maintained 13 **microservices** implementing **Domain-Driven Design (DDD)** principles with rich aggregates, value objects, and domain events for complex financial products (ISAs, GIAs, Junior ISAs), handling wealth management, trading, compliance, and account management for a production financial services platform serving mobile applications on **Google Cloud Platform (GCP)**, building resilient, scalable **distributed systems** using **.NET 8, PostgreSQL, Entity Framework Core**, and GCP services (**Cloud Run, Cloud SQL, Cloud Tasks, Secret Manager**)
- Designed and implemented an **event-driven architecture** using **Google Cloud Pub/Sub** for asynchronous communication between services, ensuring eventual consistency across distributed transactions with comprehensive error handling, retry mechanisms, and circuit breakers to achieve 99.9% uptime for critical financial operations
- Engineered complex domain models following DDD principles for financial products with sophisticated fee processing system spanning multiple services, implementing automated monthly calculations with daily compounding, discrepancy detection, and multi-tier rate hierarchies (member rates, fund rates, overrides), coordinating asset selling across portfolios with tolerance-based adjustment mechanisms
- Built JISA account management feature applying advanced domain modelling techniques for complex parent-child relationship structures, supporting multiple ownership roles (registered contact, beneficiary) with proper authorisation and compliance checks, leading major data migration from single-ownership to many-to-many ownership model using **Entity Framework Core** code-first migrations with proper indexing and constraints
- Designed mobile-first **RESTful APIs** using **FastEndpoints** with versioning strategies, backward compatibility, and mobile-optimised patterns (payload minimisation, batch operations, offline support), implementing comprehensive authentication and authorisation using **JWT Bearer tokens** with OpenIddict, supporting role-based access control across 13 services with robust API validation using FluentValidation
- Built compliance monitoring service implementing regulatory requirements, tracking deposits, withdrawals, and transactions with automated alerting for suspicious activities and threshold breaches, implementing **KYC (Know Your Customer)** workflows with document verification, risk assessment, and regulatory reporting, alongside multi-stage account holder onboarding system with third-party identity verification integration
- Integrated multiple third-party APIs including WealthKernel broker API for investment account creation, trading execution, settlement tracking, regulatory reporting and **KYC** verification workflows; Plaid (**Open Banking**) payment integration for secure bank account linking and deposit processing with webhook handling; RevenueCat for mobile subscription management and membership tier synchronisation
- Engineered automated trading system handling order execution, matching, settlement, and charge creation through WealthKernel's broker API, building portfolio rebalancing engine with sell-down and buy-up order orchestration, implementing complex financial algorithms including daily compounding fee calculations, market data handling, available units calculations, and withdrawal/deposit processing with compliance checks for regulatory limits (ISA allowances, JISA limits) and automated reconciliation
- Identified and resolved a critical double-processing bug in the shared **outbox pattern** library that had persisted for 3 years across the platform, conducting root cause analysis that revealed Entity Framework Core's optimistic concurrency model allowed race conditions during message processing, implementing a fix using **pessimistic locking** with raw SQL queries and row-level locks to ensure exactly-once message delivery semantics across all microservices
- Established comprehensive testing strategy following the testing pyramid with **xUnit, Testcontainers**, and WebApplicationFactory for **integration testing**, achieving high code coverage by building fake implementations for external services (WealthKernel, Plaid, RevenueCat, Blend) enabling fast, reliable integration tests, and implementing test data generation using AutoFixture and custom builders for complex domain objects
- Implemented **infrastructure as code** using **Pulumi** for GCP resource provisioning (Cloud Run, Cloud SQL, Pub/Sub, VPC, IAM), building **CI/CD pipelines** using **NUKE build** automation and **GitHub Actions** with environment-specific deployment strategies (development, staging, production), configuring containerised deployments with **Docker**, automated database migrations, secret management integration, and comprehensive monitoring with health checks using Google Cloud Diagnostics for observability across the distributed system
- Led platform modernisation initiatives migrating multiple services from .NET 6 to **.NET 8**, ensuring compatibility, performance improvements, and minimal downtime; refactored legacy Controller-based APIs to **FastEndpoints** pattern, improving performance, maintainability, and developer experience whilst maintaining backward compatibility for mobile clients

Present	- Led the containerisation of a .NET 6 Web API and React project with Docker, substantially improving development workflows and deployment consistency. Expertly implemented Kubernetes in Amazon EKS for application orchestration, utilising Helm charts for streamlined deployment and management of services. This strategic integration fostered a scalable, resilient hosting solution on AWS, optimising operational efficiency and minimising infrastructure overhead - Transformed database migration processes by transitioning from manual scripts to implementing FluentMigrations, automating migrations to execute upon project startup. This innovation significantly streamlined database updates, enhancing consistency and reducing manual workload - Initiated and executed a comprehensive refactoring of the API project, focusing on the efficient management of service lifetimes. This pivotal effort significantly elevated the application's performance and streamlined maintenance processes - Identified a critical SQL Server connection leak in the first week at the company, causing severe performance slowdowns for customers using SQL Server	
Oct '22 - Sep '23	Software Engineer - Dojo Ltd. - Worked alongside a compact and nimble team to create and sustain a shared service platform, streamlining the process of crafting services for in-house teams. These services were implemented in .NET 7, Kubernetes, and Docker, utilising a CQRS architectural approach and employing MongoDB - Successfully implemented an event-driven architecture on Google Cloud Platform (GCP) using PubSub, enhancing the platform's scalability and responsiveness, using a Domain-Driven Design - Participated in the upgrade initiative for all services that leverage our service, enabling them to embrace the latest version of our library and PubSub event model, and successfully transitioning all services to .NET 7. This expedited the retirement of the legacy system, resulting in a rapid and efficient transition - Part of an initiative to redesign the core internal service, consolidating it into a centralised instance that served as the primary point of communication for all other services, eliminating the need for ephemeral environments - Participated in a project to develop an in-house Identity service to replace the legacy system. Contributed to architectural decisions, integrated Auth0, and coordinated project timelines - Conducted database migration operations to modernise legacy databases by transferring historical storage data, converting it into the updated data model, and seamlessly integrating it into the new database infrastructure, thereby ensuring the synchronisation and advancement of the new database system	<i>Communications & Identity</i>
Oct '21 - Aug '22	Software Engineer - HealthHero Ltd. - Migrated old monolithic front-end repository into a monorepo solution, and rewrote the various client implementations into individual PWAs written in React - Carried out a project to introduce an adequate test framework into a .NET API named SAPI which had little-to-no testing in place, resulting in a Unit Test coverage of above 80% and an Integration Test coverage of above 70% - Contributed to a task force addressing critical bugs in a client API wrapper, leading to a 2-year contract renewal - Contributed to the development of a specialised database migration tool tailored for Azure CosmosDB, enabling the anonymisation of personally identifiable information (PII) data to ensure compliance with GDPR regulations while facilitating data analysis capabilities for cross-functional teams	<i>MedSol PWA / SAPI</i>
Jul '19 - Oct '21	Junior Software Engineer - Argus Media Ltd. - Engaged in a greenfield project tasked with the creation of a highly customisable user workspace, designed to offer critical price benchmark data. Developed utilising Angular and .NET Core, the project incorporated a robust database architecture, leveraging both Amazon DynamoDB and SQL Server - Implemented the Noda Time API to overcome the various design limitations in the .NET Time API - Carried out a project to introduce a theming service to asynchronously load and cache selected custom themes for the user, resulting in clients opting to use non-default themes at a rate of 84% - Attending monthly security advocacy meetings to identify areas of vulnerability within the code base and infrastructure DevOps pipeline - Collaborated on a greenfield project using AWS API Gateway and serverless computing to implement company-built custom workspaces	<i>Direct Workspaces</i>
Jun '18 -	QA Engineer - ASOS PLC	<i>Bag API</i>

Jun '19	- Implemented new features to improve the user experience on an Azure hosted C# Facebook Chatbot - Developed and implemented a stored procedure to automatically remove test-user entries from CosmosDB during each deployment - Introduced brand mapping functionality so end-users could specify specific brands within their searches, thus allowing users to see what they actually want to see - Attended twice-weekly code craftsmanship classes run by the Lead Developer in order to make the transfer over to Software Engineer - Completely refactored existing test suite left behind by prior Senior QA to make it more robust and mapped according to the Testing Pyramid. Resulted in reduced deployment times of around 40% and a less "flaky" test suite
May '17 - Jun '18	Associate Software Developer in Test - Veritas LLC <i>Information Classifier</i> - Implemented new features on a Java microservice for data discovery and visualisation. An NLP solution implementing Stanford NLP, using a TDD and BDD approach to development - Created an internal performance monitoring dashboard using Chart.js to monitor any degradation in the classification engine - Writing Unit, Integration, and UI tests using Mockito, TestNG, and Protractor for the End-to-End tests - Configured and remastered build & release pipelines with CI/CD using Maven, Jenkins and JIRA

Education

2012-2017	Bachelor of Science in Computer Science Brunel University , London	Award: Second Upper Class (2:1)
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About Me

- Programming and learning new technologies has now become a big part of my life, and I try to keep myself up-to-date with the latest in tech by attending conferences, joining meetup groups, and going to training courses
- I am an avid practitioner of Muay Thai and Boxing. So if you see me pop in with black eyes and several bruises, don't be alarmed!
- I like to keep fit and help others reach their fitness goals. So far I have lost approximately 36kg and have no plans to stop yet! I have taught many friends and family about the best (and healthiest) ways to reach their goals, whether they are trying to gain mass or lose weight, with successful results
- I maintain a personal blog at harveysingh.xyz, built with Astro and deployed on Cloudflare. The site serves as a platform where I document my journey as a developer, share insights about my career, and explore interesting technologies and projects I've worked with along the way

References available upon request